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April 24, 1967
SUPERSEDING
Fed. Spec. UU-W-101a
July 3, 1935

FEDERAL SPECIFICATION

BUILDING BOARD, HARD PRESSED, VEGETABLE FIBER (LAMINATED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a vegetable-base fiber, medium density, building board.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-S-781 - Steel, Strapping, Flat.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
PPP-C-650 - Crates, Wood, Open and Covered.
PPP-T-60 - Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subcription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The board shall be a solid rectangular panel.

3.2 Material. The board shall be composed of vegetable fibers and a water-resistant size. The size may be wax, resin, or any other additive which will increase water resistance.

3.3 Density. The board shall have a maximum density of not more than 45 pounds per cubic foot.

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3.4 Dimensions. Unless otherwise specified (see 6.2), boards shall be 4 feet wide and 8 feet long. Nominal board thicknesses shall be 3/16, 1/4, 5/16, or 3/8 inch, as specified (see 6.2).

3.4.1 Tolerances. The tolerance of the width and length of the board shall be minus 1/8, plus 1/16 inch. Thickness tolerance up to 5/16 inch shall be plus or minus 5 percent, and 3/8 inch plus or minus 10 percent.

3.5 Flexural strength. The average flexural strength of the board shall be not less than the value shown in table I for the applicable board nominal thickness.

TABLE I. Flexural strength

Nominal thickness	Flexural strength, minimum
(Inch)	(Pounds)
3/16	7
1/4	12
5/16	18
3/8	27

3.6 Linear expansion. The average linear expansion of the board shall be not more than 0.75 percent after exposure to a 50 percent increase in the relative humidity.

3.7 Finish. One face of the board shall be flat, unblemished, and coated with a varnish. The obverse face of the board shall be coated with a vanish or equally water-resistant protective coating, and may be either smooth or embossed.

3.8 Workmanship. The board shall be clean, have sound, square edges, and shall be free of extraneous chips, curls, flakes, or shavings. Corners shall be square.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.1.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.3.2 Examination. Samples selected in accordance with 4.3.1.1, shall be examined as specified in 4.4.1. AQL shall be 4.0 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1.2, shall be tested as specified in 4.4.2. Failure of any test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The board shall be examined for the following defects:

101. Materials not as specified.

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102. Finish not as specified.
103. Workmanship not as specified.

4.4.2 Test Methods.

4.4.2.1 Test conditions. Specimens of the size specified for each test shall be cut from each sample board. Specimens shall be conditioned prior to testing to a constant weight and moisture content in a conditioning chamber maintained at a relative humidity of 65 ± 1 percent and a temperature of 70° to 75° F.

4.4.2.2 Density. Specimen size shall be 12 by 12 inches. Weigh each specimen and calculate the weight per cubic foot (density exceeding 45 pounds per cubic foot shall constitute failure of this test.

4.4.2.3 Flexural strength. Specimen size shall be 18 inches long by 3 inches wide. Place the specimen on horizontal, paralleled supports 12 inches apart, and apply the load at midspan on a bearing parallel to the end supports so that deflection takes place at a rate of 12 ± 2 inches per minute, until the specimen breaks. The bearing and supports shall be rounded to a radius of approximately 3/8 inch. A total of six specimens shall be tested from each sample board. Three specimens shall be cut parallel to the board length and three at right angles thereto. An average flexural strength which is less than the value specified in table I shall constitute failure of this test.

4.4.2.4 Linear expansion. Six specimens, 12 inches long by 3 inches wide, shall be cut from each sample board. Three specimens shall be cut parallel to the length of the board and three at right angles thereto. Coat two small areas 10 inches apart with cryon or marking pencil. Within these prepared areas, scribe reference marks exactly 10 inches apart with a fine pointed awl or blade. Then expose the specimens to an atmosphere of 70° to 75° F. at 97 percent relative humidity for 24 hours. Remeasure the distance between the scribe marks and compute the increase in distance as a percent of the original distance. Measurements shall be made to the nearest 0.005 inch. An increase in distance between the scribe marks on any specimen, after exposure to the test atmosphere, which exceeds 0.75 percent shall constitute failure of this test.

4.4.2.5 Dimensions. Measure the width and length of the board at each end and at midwidth and midlength. Measure the thickness at each corner and at the center of the board with a micrometer accurate to 0.001 inch. Any measurement exceeding the tolerance specified in 3.4.1 shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For purposes of inspection, the unit of product shall be a completed pack prepared for shipment.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

104. Materials and containers not as specified for level A or B.
105. Building board not wrapped with waterproof paper for level A.
106. Strapping not zinc coated for level A.
107. Building board not strapped as specified for level B.
108. Marking illegible or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Not applicable.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Building board, in quantities not to exceed the following, shall be packed in crates conforming to PPP-C-650, style A, type II, except the crate shall be modified by four 2 by 4 inch skids and the 1,000 pound weight limit shall be waived.

3/16 inch thick, 88 sheets.
1/4 inch thick, 60 sheets.
5/16 inch thick, 52 sheets.
3/8 inch thick, 44 sheets.

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Boards of like description shall be carefully and uniformly piled. Each pile shall contain a quantity of sheets most convenient for handling and shall be wrapped with waterproof paper conforming to PPP-B-1055, class E-1, or L-4. The paper shall be sealed with tape conforming to PPP-T-60, class 1, or heat sealed, as applicable. Quantities and sizes indicated above shall be packed in each crate. Sufficient dunnage shall be placed within the crate to protect corners and edges of boards from damage. The crate shall be closed and strapped in accordance with the appendix to the crate specification. Strapping shall be zinc coated.

5.2.2 Level B. Building board, in quantities as specified in 5.2.1, shall be secured in bundles with steel strapping conforming to QQ-S-781, type 1, class A. Plywood waster sheets, 1/4-inch thick, shall be placed on the top and bottom of the bundle prior to application of strapping. Strapping shall consist of two straps, each located 2 feet from the ends of the bundle, and placed girthwise around the bundle in a lateral direction. A third strap shall be placed girthwise in the center of the bundle in a longitudinal direction. Strapping shall be 5/8-inch wide by 0.023 inch thick.

5.2.3 Level C. The boards shall be packed to insure arrival at destination in satisfactory condition in a container complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special markings required by the contract or order, the bundles and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The board covered by this specification is intended for use as a wall covering, target backing, map or photo mounting, displays, and in packaging and packing operations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Dimensions of board required if other than as specified in 3.4.
(c) Thickness of board required (see 3.4).
(d) Selection of applicable level of packing required (see 5.2).

Level B packing is intended to provide economical but limited protection and should be specified only when it is determined that the building board will be held in covered storage.

MILITARY INTEREST:

Custodians:

Army - ME
Navy - YD
Air Force - 69

Review activities:

Navy - YD
Air Force - 69

User activities:

Army - MI
Navy - SH

Other:

NSA

Preparing activity:

Army - ME

Code "C"

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CIVIL AGENCY INTEREST:

AGR HEW
DC INT VA
GSA JUS

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

Orders for this publication are to be placed with General Serviced Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies of documents referenced herein. Price 5 cents each.

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